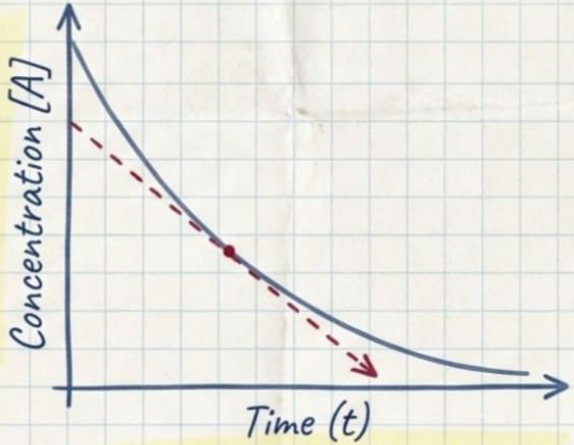
Chemical Kinetics



A Master Study Guide: Rates, Mechanisms, and Molecular Collisions.

Topics: Rate Laws, Arrhenius Theory, Molecular Collisions.

Average rate & instantaneous rate

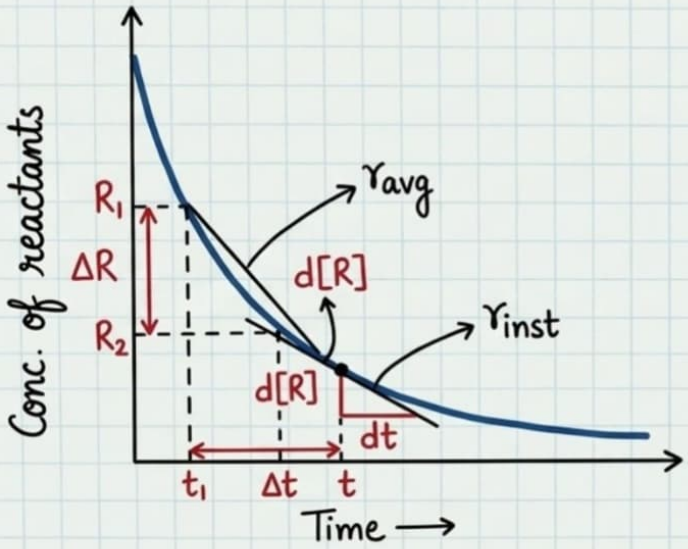
1. $\text{Rate}_{(avg)} = \dfrac{-\Delta[R]}{\Delta t}$ → (It can be defined as the change in the concentration of reactant/product per unit time).

$\text{Rate}_{(inst)} = \dfrac{-d[R]}{dt}$

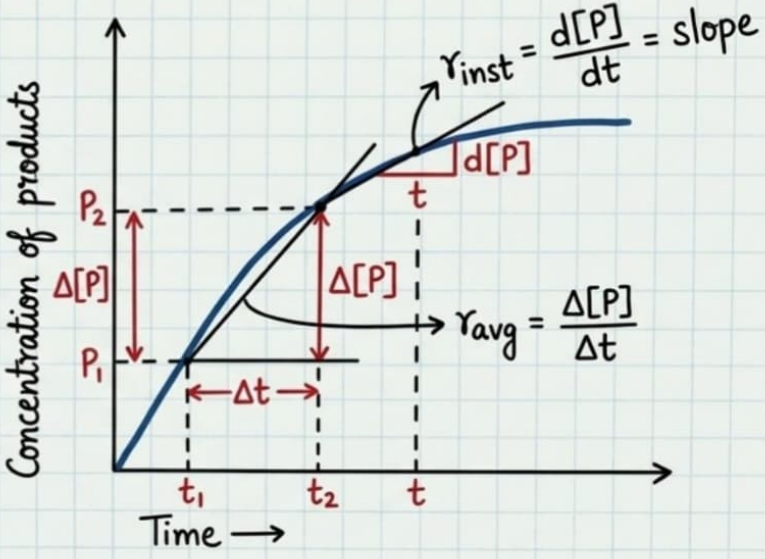
Δ = macroscopic interval,
d = infinitesimal instant.

-ve sign shows that the slope of [R] v/s t will go downwards ∴ with time, conc. of reactants decrease.

Rate Matrix: Instantaneous & Average Rates



Instantaneous & avg rate of reactants



Instantaneous & avg rate of products.

Rate of reactions (with stoichiometric coefficients)

For expressing the rate of such a rxn where stoichiometric coefficients of reactants or products are not equal to one, rate of disappearance of any of the reactants or rate of appearance of products is divided by their respective coefficient.

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1. $2HI(g) \longrightarrow H_2(g) + I_2(g)$

$$\text{Rate} = -\frac{1}{2}\left(\frac{\Delta[HI]}{\Delta t}\right) = \frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[I_2]}{\Delta t}$$

E.g: 200 mol HI $\longrightarrow$ 100 mol H_2 + 100 mol I_2

$$\text{Rate} = -\frac{1}{2}\left(\frac{\Delta[HI]}{\Delta t}\right) = 100 \qquad \frac{\Delta[HI]}{\Delta t} = 2R \text{ (rate of rxn of HI)}$$

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2. $2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$

Rate of rxn: $R = -\frac{1}{2}\left(\frac{\Delta[N_2O_5]}{\Delta t}\right) = \frac{1}{4}\left(\frac{\Delta[NO_2]}{\Delta t}\right) = \frac{\Delta[O_2]}{\Delta t}$

Rate of consumption: $R = \frac{\Delta[N_2O_5]}{\Delta t} = 2R$

Rate of formation: $R = \frac{\Delta[NO_2]}{\Delta t} = 4R$; $R = \frac{\Delta[O_2]}{\Delta t}$

Factors influencing rate of reaction: Rate law & Order

- **Rate Law**

The representation of rate of rxn in terms of concentration of the reactants is called as rate law.

1. Rate of a rxn decreases with time as the concentration of reactants decreases. Thus, rate of reaction depends upon the concentration of reactants.

- Rate law is the expression in which rxn rate is given in terms of molar concentration of reactants with each term raised to some power, which may or may not be the same as stoichiometric coefficients.

$$aA + bB \longrightarrow cC + dD$$

$$\text{Rate} \propto [A]^x[B]^y \quad \text{or} \quad \text{Rate} = k[A]^x[B]^y \Rightarrow \frac{-d[R]}{dt} = k[A]^x[B]^y$$

- **Order**

Powers of the concentration of the reactants in the rate law expression is called the order of that chemical reaction.

1. Reactions taking place in one step are called as elementary reactions.
2. Reactions taking place in more than one steps are called as complex reactions.

Units:

Unit = $(\text{mol L}^{-1})^{1-n}\ \text{s}^{-1}$

where n = order of rxn.

0 order rxn = $\text{mol L}^{-1}\ \text{s}^{-1}$

1st order rxn = s^{-1}

2nd order rxn = $\text{L mol}^{-1}\ \text{s}^{-1}$

Experimental Determination: $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

Rate of rxn $= \frac{1}{2}x = \frac{x}{2}$. $\therefore$ rate of consumption of $O_2 = x$ & $NO = \frac{x}{2}$.

Lab Data

	Initial [NO]	Initial $[O_2]$	Initial rate of $[NO_2]$
1.	0·3	0·3	0·096
2.	0·6	0·3	0·384
3.	0·3	0·6	0·192

Now, suppose:- $R = k[NO]^x[O_2]^y$

$$R = \frac{k\cancel{(0\cdot3)^x}(0\cdot3)^y}{\cancel{k(0\cdot6)^x}(0\cdot3)^y} = \frac{0\cdot096}{0\cdot384}$$

$$\Rightarrow \frac{(0\cdot3)^x}{(0\cdot6)^x} = \frac{1}{4}$$

$$\Rightarrow \left(\frac{1}{2}\right)^x = \left(\frac{1}{2}\right)^2 \quad \boxed{\Rightarrow x = 2}$$

Similarly, $R = \frac{k\cancel{(0\cdot3)^x}(0\cdot3)^y}{\cancel{k(0\cdot3)^x}(0\cdot6)^y} = \frac{0\cdot096}{0\cdot192}$

$$\Rightarrow \frac{(0\cdot3)^y}{(0\cdot6)^y} = \frac{1}{2}$$

$$\Rightarrow \left(\frac{1}{2}\right)^y = \left(\frac{1}{2}\right)^1 \quad \boxed{\Rightarrow y = 1}$$

$$\therefore \frac{-d[R]}{dt} = k[NO]^2[O_2]$$

Molecularity of a reaction

The no. of reacting species (atoms, ions or molecules) taking part in an elementary reaction, which must collide simultaneously in order to bring about a chemical rxn is called its molecularity.

1. Unimolecular: $NH_4NO_2 \longrightarrow N_2 + 2H_2O$
   In unimolecular rxns, a single compound gets decomposed into 2 or more compounds.

2. Bimolecular: $2HI \longrightarrow H_2 + I_2$
   In bimolecular rxns, 2 compounds collide with each other.

3. Trimolecular: $O_2 + 2NO \longrightarrow 2NO_2$
   In trimolecular or tetramolecular rxns, 3 or more compounds collide simultaneously.

Simultaneous collision of >3 molecules is very rare.

Rate determining step or slow step

1. Overall rate of a rxn, in a complex rxn mechanism, depends upon the slow step or rate determining step of the rxn.

2. E.g: Decomposition of H_2O_2:-

$$2H_2O_2 \xrightarrow{I^- \text{ Alkaline medium}} 2H_2O + O_2$$

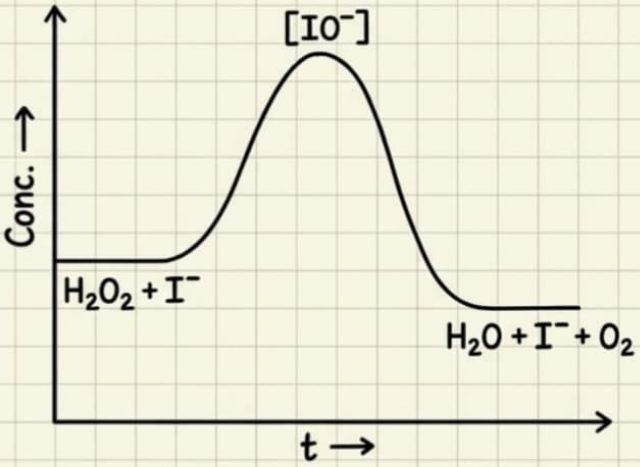
Its rate equation is found as,

$$\text{Rate} = \frac{-d[H_2O_2]}{dt} = k[H_2O_2][I^-]$$

This rxn is 1st order & takes place in 2 steps:-

(1) $H_2O_2 + I^- \longrightarrow H_2O + IO^-$ slow step / Intermediate

(2) $H_2O_2 + IO^- \longrightarrow H_2O + I^- + O_2$



3. Species like IO^- don't exist much and are called as intermediates, as it is formed during the course of rxn but does not remain till the end of the rxn.

Conclusions: Order vs. Molecularity

ORDER	MOLECULARITY
- Order is an experimental qty.	- Theoretical concept based on mechanism.
- Can be 0 or fractional.	- Molecularity can neither be 0 nor fractional.
- Order is applicable for elementary & complex rxns both.	- Molecularity is applicable for complex rxns.*

*elementary steps within them

Integrated rate equations: Zero order reactions

Rate of rxn refers to only those particles that come in contact with the surface of catalyst to react faster and get converted into product.

For rxn, $R \rightarrow P$

$$\text{Rate} = -\frac{d[R]}{dt} = k[R]^0$$

or $$\frac{-d[R]}{[R]^0} = k\,dt \Rightarrow d[R] = -k\,dt$$

$$\Big\} \because \int dx = x$$

Integrating b/s,

$$\int_{R_0}^{R} d[R] = \int_0^t -k\,dt$$

$$[R]_{R_0}^{R} = -k[t]_0^t$$

$$[R] - [R_0] = -kt$$

$$\boxed{\therefore [R] = -kt + [R_0]}$$

Half-life ($t_{1/2}$)

1. When, $R = \frac{R_0}{2}$, this condition is called as half-life.

Substituting into zero-order eq:

$$\frac{R_0}{2} = -kt_{1/2} + R_0$$

$$kt_{1/2} = R_0 - \frac{R_0}{2}$$

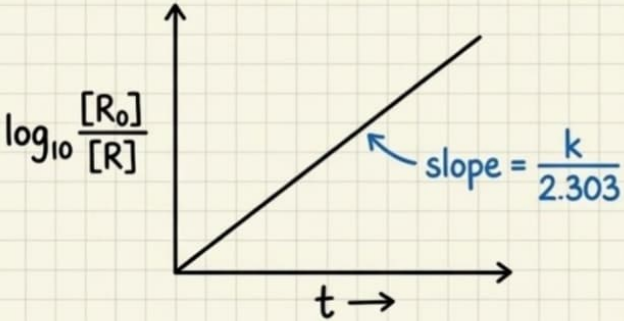
$$kt_{1/2} = \frac{R_0}{2}$$

$$\boxed{\therefore t_{1/2} = \frac{R_0}{2k}}$$

Expert Note: For a zero-order reaction, half-life is directly proportional to initial concentration $[R_0]$.

First order rxn: Graphical Analysis

Log Graph

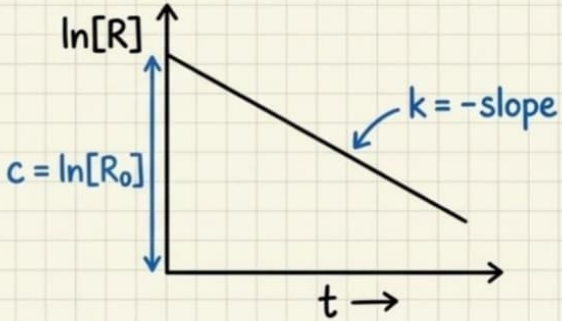


Graph b/w $\log\frac{[R_0]}{[R]}$ v/s t for 1st order

$$2.303 \times \log_{10}\left(\frac{[R_0]}{[R]}\right) = kt \Rightarrow \log_{10}\left(\frac{[R_0]}{[R]}\right) = \frac{kt}{2.303}$$

$$y = mx \quad \left(\text{where } y = \frac{\log_{10}[R_0]}{[R]},\ m = \frac{k}{2.303},\ x = t\right)$$

Ln Graph



Graph b/w $\ln[R]$ & t

$$\ln\left(\frac{[R_0]}{[R]}\right) = kt \Rightarrow \ln[R_0] - \ln[R] = kt$$

$$\Rightarrow -\ln[R] = kt - \ln[R_0]$$

$$y = mx + c \quad (\text{where } y = -\ln[R],\ m = kt,\ c = \ln[R_0]).$$

Thus, $k = -\text{slope}$.

First order gas phase rxn

$$2N_2O_{5(g)} \longrightarrow 2N_2O_{4(g)} + O_{2(g)}$$

At t=0 (Initial pressure)	P	0	0
At t=100s (Final pressure)	P − x	x	$\frac{x}{2} = 0.512$

$$P - x + x + \frac{x}{2} = 0.512$$

$$P + \frac{x}{2} = 0.512 \text{ atm}$$

Let $P = 0.5$ atm. $\therefore\ 0.5 + \frac{x}{2} = 0.512 \Rightarrow \frac{x}{2} = 0.512 - 0.500 = 0.012 \Rightarrow x = 0.024$

$$P_{N_2O_4} = 0.024 \ \ \& \ \ P_{N_2O_5} = 0.5 - 0.024 = 0.476 \text{ atm}$$

$$2.303 \log_{10}\left(\frac{P_{initial}}{P_{final}}\right) = kt$$

$$k = \frac{2.303}{100} \log\left(\frac{0.5}{0.476}\right) = 4.98 \times 10^{-4}\ s^{-1}$$

$$\boxed{k = 4.98 \times 10^{-4}\ s^{-1}}$$

Temperature dependence of the rate of a reaction

1. For a chemical rxn, with rise in temperature by 10°, the rate constant is nearly doubled.
2. It was first proposed by Dutch scientist J.H. van't Hoff but Swedish chemist Arrhenius provided its physical justification.

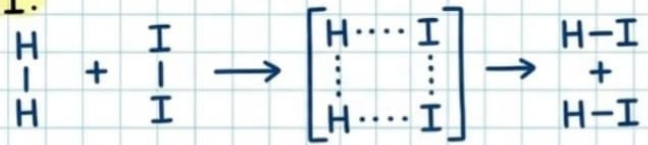
$$k = Ae^{-E_a/RT}$$

Where A is the Arrhenius factor or frequency factor, R is gas constant & T is temperature.

Arrhenius Theory & The Activated Complex

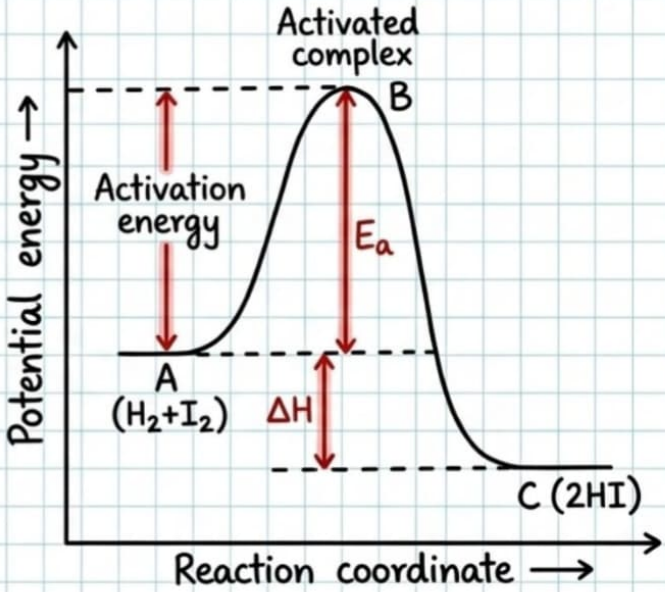
1. Acc to Arrhenius, this rxn takes place only when a molecule of hydrogen and a molecule of iodine collide to form an unstable intermediate.

2. It exists for a very short time & then breaks up to form 2 molecule HI.



3. The energy required to form this intermediate, called activated complex, is called as activation Energy (E_a).

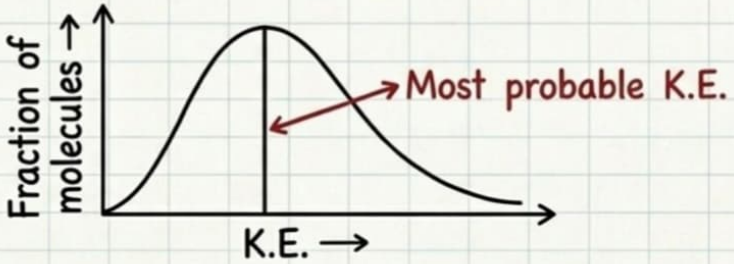
4. Reaction coordinate represents the profile of energy change when reactants change into products.



Maxwell-Boltzmann Statistics

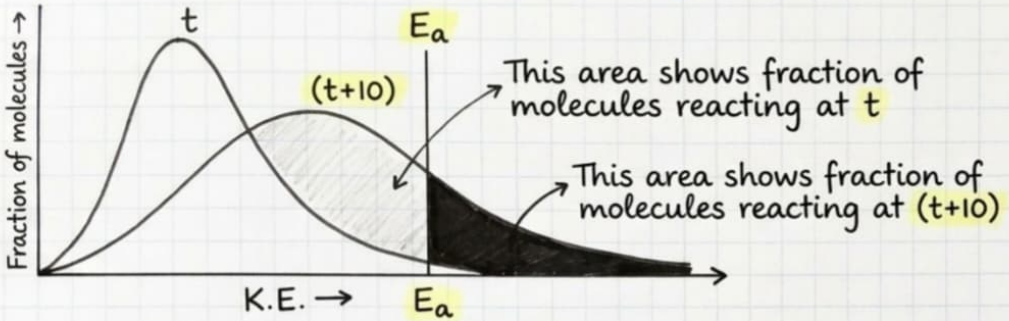
5. In the molecules, the reacting species do not have the same K.E.
6. Ludwig Boltzmann & James Clark Maxwell used statistics to predict the behavior of large no. of molecules.
7. According to them, the distt of K.E may be described by plotting the fraction of molecules (N_E/N_T) v/s K.E.

   Here, $N_E \rightarrow$ No. of molecules with energy E.
   $N_T \rightarrow$ Total no. of molecules.



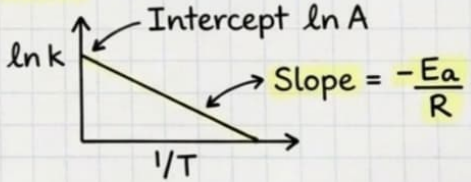
8. The peak of the curve corresponds to the most probable K.E i.e, K.E of maximum fraction of the molecules with energies higher or lower than this value.

The 10-Degree Shift & Activation Energy



9. As the temperature is raised, the maximum of the curve moves to the higher energy value and the curve broadens out...
11. It is clear from the diagram that in the curve at (t+10), the area showing the fraction of molecules having energy equal to or greater than activation energy gets doubled leading to double the rate of a rxn.

12. $$\ln k = -\frac{E_a}{RT} + \ln A$$



13. Increasing the temperature or decreasing the activation energy will result in an increase in the in the rate of reaction and an exponential increase in the rate constant.

Arrhenius Equation at Two Temperatures

$$\ln K_1 = \frac{-E_a}{RT_1} + \ln A \quad \text{(For temp } T_1\text{)}$$

Similarly, $$\ln K_2 = \frac{-E_a}{RT_2} + \ln A \quad \text{(For temp } T_2\text{)}$$

$$\therefore \ln K_2 - \ln K_1 = \left(\frac{-E_a}{RT_2} + \ln A\right) - \left(\frac{-E_a}{RT_1} + \ln A\right)$$

$$\ln K_2 - \ln K_1 = \frac{E_a}{RT_1} - \frac{E_a}{RT_2}$$

or $$\ln\left(\frac{K_2}{K_1}\right) = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Converting to base 10

or $$\log\left(\frac{K_2}{K_1}\right) = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

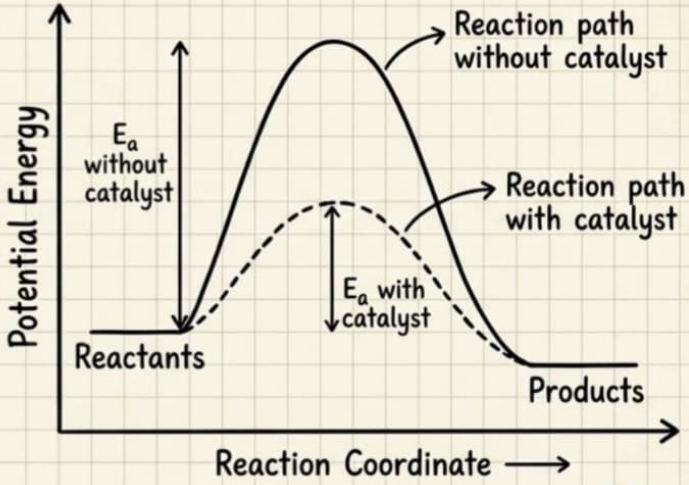
or $$\boxed{\log\left(\frac{K_2}{K_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]}$$

Effect of Catalyst & Intermediate Complex Theory

1. A catalyst is a substance which increases the rate of a reaction without itself undergoing any permanent chemical change. (E.g: MnO_2 catalyses to increase its rate considerably).
2. Those substance that reduces the rate of rxn are called inhibitor.

✱ <u>Intermediate complex theory</u> :-

1. Acc to this theory, a catalyst participates in a chemical rxn by forming temporary bonds with the reactants resulting in an intermediate complex.
2. This has transitory existence and decomposes to yield products and the catalyst.



<u>Expert Note</u>: Catalyst provides an alternate pathway by reducing E_a. It neither changes ΔG nor K, but helps attain equilibrium faster.

Collision theory of chemical rxns

1. It was developed by Max Trautz & William Lewis in 1916-1918.
2. It is based on kinetic theory of gases. Acc to this theory, the reactant molecules assumed to be hard spheres and reaction is postulated to occur when molecules collide with each other.
3. The no. of collisions per second per unit volume of the reaction mixture is known as collision frequency (Z).

$$\boxed{\therefore \text{Rate} = Z_{AB} e^{-E_a/RT}}$$

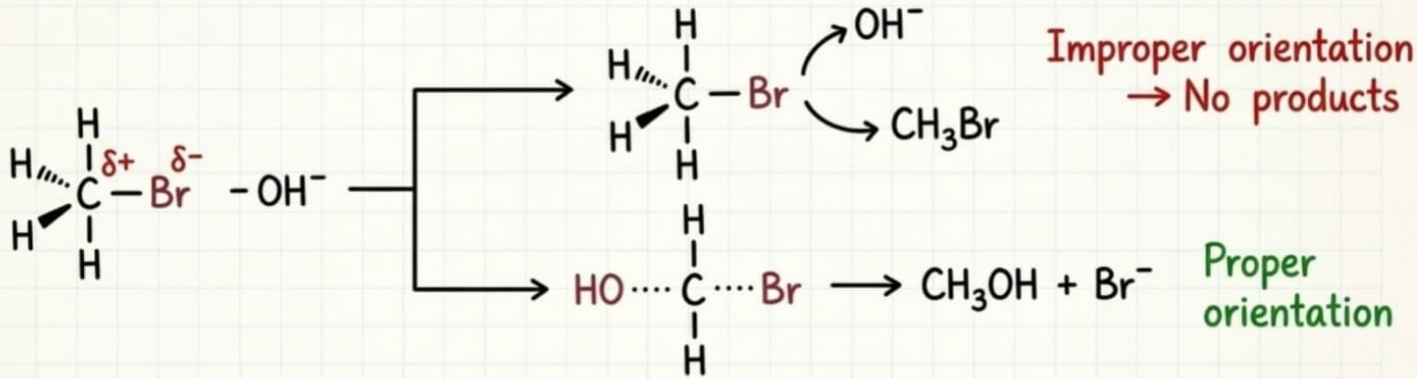
where, Z_{AB} is the collision frequency of reactants A & B.

* <u>Orientation of molecules</u>:-

1. For complex molecules significant deviations are observed. This is because all collisions do not lead to the formation of products.
2. The collisions in which molecules collide with sufficient K.E (called threshold energy) and proper orientation so as to facilitate breaking of bonds...

Orientation of Molecules & The Steric Factor

$CH_3Br + OH^-$ depends upon the orientation of reactants molecules:



3. Thus, proper orientation of reactants molecules lead to product formation whereas improper orientation makes them simply bounce back and no products are formed.

4. To account for effective collision another factor P, called probability or steric factor was introduced.